

Continuous functions – 6

Functional equations under a regularity assumption

1 сентября 2026 года

Theoretical minimum.

Theorem 1 (Cauchy). If $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(x+y) = f(x) + f(y)$ and is continuous at a single point, or bounded on some interval, or measurable, then $f(x) = cx$. Without any regularity there are, by the axiom of choice, discontinuous solutions: they are the $\mathbb{Q}$ -linear maps built from a Hamel basis, and their graphs are dense in the plane.

Theorem 2 (the four classical equations). Under continuity, $f(x+y) = f(x)f(y)$ gives $f = e^{cx}$ or $f \equiv 0$; $f(xy) = f(x) + f(y)$ on $(0, \infty)$ gives $f = c \ln x$; $f(xy) = f(x)f(y)$ gives $f = x^c$ or $f \equiv 0$; and Jensen's equation $f\left(\frac{x+y}{2}\right) = \frac{f(x)+f(y)}{2}$ gives $f = ax + b$.

Fact 1. The working scheme: substitute special values to obtain $f(0)$, $f(1)$ and symmetry; prove the equation for integers, then for rationals; and only at the last step invoke continuity to pass from $\mathbb{Q}$ to $\mathbb{R}$.

1. Prove Theorem 1 in all three regularity versions, and prove that a discontinuous additive function has a graph dense in the plane.
2. Deduce the four classical equations of Theorem 2 from the Cauchy equation, being explicit about where continuity is used.
3. (Putnam 2022) Find all continuous $f: \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that $f(xf(y)) + f(yf(x)) = 1 + f(x+y)$ for all $x, y > 0$.
4. (IMS 2016) Find all functions $f: (0, +\infty) \rightarrow (0, +\infty)$ satisfying $f(xf(y)) = yf(x)$ for all $x, y > 0$ together with $\lim_{x \rightarrow +\infty} f(x) = 0$.
5. (RMO) Find all continuous $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $n^2 \int_x^{x+1/n} f(t) dt = nf(x) + \frac{1}{2}$ for every $x \in \mathbb{R}$ and every positive integer n .